Corporate Financial & Margin Performance Dashboard

Power BI Internship Project Report

This report documents the proposed dashboard design, data preparation, financial measures, visuals, scenario modelling, and validation steps for the internship task.

PROJECT ITEM	DESCRIPTION
Business focus	Revenue, discounts, COGS, profit, margin and segment performance
Tools	Power BI Desktop, Power Query, DAX and What-if parameters
Source data	Cleaned Microsoft Financial Sample CSV supplied for the project
Proposed report	One PBIX file containing four interactive pages

Scope note: The supplied CSV does not include a budget target field or a complete operating-expense breakdown. Actual-versus-budget variance and operating margin must not be reported as verified values unless suitable additional data is provided.

This is an implementation report, not a claim that the dashboard has already been built or that financial findings have already been validated.

1. Executive Summary

The Corporate Financial & Margin Performance Dashboard is designed to explain how gross sales become net sales and profit, and how performance differs across months, countries, products, and business segments. The recommended report has four pages to cover the internship brief while keeping visuals organised and readable.

Project objectives

• Present Gross Sales, Discounts, Net Sales, COGS, Profit, and Profit Margin as clear KPI cards.

• Track monthly financial performance and compare segments, products, and countries.

• Use a waterfall chart to show the impact of discounts and COGS on profit.

• Add interactive discount and cost scenario sliders to estimate the effect on profit.

• Include actual-versus-budget variance only when valid budget data is available.

Recommended page structure

PAGE	TITLE	PURPOSE
1	Financial Overview	Headline KPIs, monthly sales/profit trend, sales by segment and profit by country
2	Revenue & Margin Analysis	Revenue waterfall, discounts, COGS ratio and margin trends
3	Segment & Product Performance	Country, segment and product profitability comparisons
4	Scenario Planning & Variance	What-if modelling and budget variance if approved targets exist

Expected value

The completed report should help users identify major sales and profit contributors, recognise products with comparatively weak margins, understand the effect of discounting and cost changes, and discuss potential scenarios. Business conclusions should be written only after reviewing validated dashboard outputs.

2. Data Source and Preparation

The cleaned financial sample CSV is described as containing Segment, Country, Product, Discount Band, Units Sold, Manufacturing Price, Sale Price, Gross Sales, Discounts, Sales, COGS, Profit, Date, Month Number, Month Name, and Year. Verify the exact field names in Power Query before copying the DAX below.

Power Query workflow

• Choose Home → Get data → Text/CSV and import the cleaned CSV.

• Select Transform Data. Set Date as Date; Year and Month Number as whole numbers; category fields as text; and financial fields as numeric.

• Check blanks, errors, duplicate rows, unexpected negative values, and numeric fields imported as text.

• Validate whether Sales equals Gross Sales minus Discounts and whether Profit equals Sales minus COGS. Investigate differences instead of silently changing values.

• Select Close & Apply and save the report as a PBIX file.

Create a date table

```
DateTable = ADDCOLUMNS( CALENDAR(MIN(Financials[Date]), MAX(Financials[Date])), "Year", YEAR([Date]),
"Month Number", MONTH([Date]), "Month Name", FORMAT([Date], "MMMM"), "Year Month", FORMAT([Date],
"yyyy-MM"), "Quarter", "Q" & FORMAT([Date], "Q") )
```

Mark DateTable as the date table, sort Month Name by Month Number, and create a one-to-many relationship from DateTable[Date] to Financials[Date]. The code assumes the imported data table is named Financials; adjust names if your model differs.

Important data limitations

LIMITATION	HOW TO HANDLE IT
No budget field	Request an approved budget table. Do not label assumed targets as actual budget.
Country is available; separate region/cost-centre fields are not confirmed	Use Country for geographic comparisons. Create a region mapping only if it is documented and valid.
No full operating-expense breakdown	Report the supplied Profit and Profit Margin; do not call it Operating Margin without supporting data.
Scenario values are user-defined assumptions	Label results as scenario estimates, not guaranteed forecasts.

3. Financial Measures and DAX

Create measures using Modeling → New measure. The formulas use the field names described for the supplied CSV; update references if your actual model uses different names.

Core financial measures

```
Total Gross Sales = SUM(Financials[Gross Sales]) Total Discounts = SUM(Financials[Discounts]) Total Sales = SUM(Financials[Sales]) Total COGS = SUM(Financials[COGS]) Total Profit = SUM(Financials[Profit]) Total Units Sold = SUM(Financials[Units Sold])
```

Margin and ratio measures

```
Gross Profit % = DIVIDE([Total Profit], [Total Sales], 0) COGS Ratio = DIVIDE([Total COGS], [Total Sales], 0) Discount Rate = DIVIDE([Total Discounts], [Total Gross Sales], 0) Segment Margin % = DIVIDE([Total Profit], [Total Sales], 0) Average Selling Price = DIVIDE([Total Sales], [Total Units Sold], 0)
```

Format percentage measures as percentages with two decimal places. Format financial measures consistently using the reporting currency agreed for the dataset. Gross Profit % is calculated from the source Profit field; it should not be called operating margin.

Reconciliation checks

```
Sales Reconciliation Difference = [Total Gross Sales] - [Total Discounts] - [Total Sales] Profit Reconciliation Difference = [Total Sales] - [Total COGS] - [Total Profit]
```

Check these measures at the total level and across segments/products. Zero indicates the expected arithmetic reconciles in the current filter context. Non-zero results may reflect rounding, source definitions, or data-quality issues and should be investigated.

MEASURE	INTERPRETATION
Total Gross Sales	Sales before discounts as represented in the source data
Total Discounts	Discount amount recorded in the source data
Total Sales	Net sales field supplied by the source
Total COGS	Cost of goods sold
Total Profit	Profit field supplied by the source
Gross Profit %	Total Profit divided by Total Sales
COGS Ratio	Total COGS divided by Total Sales
Discount Rate	Total Discounts divided by Total Gross Sales

4. Page Design and Visual Plan

Page 1 — Financial Overview

Purpose: Give the viewer a quick understanding of the financial position and trends.

VISUAL	FIELDS / MEASURES	PURPOSE
Six KPI cards	Total Gross Sales, Total Sales, Total Profit, Total Discounts, Total COGS, Gross Profit %	Headline performance
Line chart	Axis: DateTable[Year Month]; values: Total Sales and Total Profit	Monthly trend
Bar chart	Segment; Total Sales	Sales by segment
Bar chart	Country; Total Profit	Profit by country
Slicers	Year, Country, Segment	Interactive filtering

Page 2 — Revenue & Margin Analysis

Purpose: Explain how discounts and COGS reduce sales available for profit, and examine margin patterns.

VISUAL	FIELDS / MEASURES	PURPOSE
KPI cards	Discount Rate, COGS Ratio, Gross Profit %	Efficiency indicators
Waterfall chart	Gross Sales → Discounts → Net Sales → COGS → Profit	Revenue-to-profit bridge
Bar chart	Product and Total Discounts	Discount distribution by product
Line chart	Year Month and Gross Profit %	Margin trend
Column chart	Year Month; Total Sales and Total COGS	Sales versus costs
Slicers	Year, Product, Discount Band	Focused analysis

Waterfall note

Create a disconnected stage table and a SWITCH measure that returns Gross Sales as positive, Discounts as negative, Net Sales as a total, COGS as negative, and Profit as a total. In the native Waterfall visual, mark Net Sales and Profit as totals where supported. Verify the displayed bridge against the source data.

4. Page Design and Visual Plan (continued)

Page 3 — Segment & Product Performance

Purpose: Compare profitability across products, segments, and countries.

VISUAL	FIELDS / MEASURES	PURPOSE
Bar chart	Segment; Total Sales	Segment sales ranking
Bar chart	Product; Total Profit	Product profit ranking
Scatter chart	X: Total Sales; Y: Total Profit; product-level detail; optional Segment legend	Find high-sales/low-profit patterns
Bar chart	Country; Total Profit	Country profitability
Matrix	Rows: Product; Columns: Segment; Values: Total Sales, Total Profit, Gross Profit %	Detailed comparison
Slicers	Year, Country, Segment, Product	Explore selected categories

Page 4 — Scenario Planning & Variance

Purpose: Let users test hypothetical discount and cost changes and see their impact on estimated profit.

VISUAL	CONFIGURATION	PURPOSE
What-if parameter	Discount Rate Adjustment: 0%–30%, 1% step	Assumed discount rate
What-if parameter	COGS Change %: –20% to +20%, 1% step	Assumed cost change
KPI cards	Scenario Net Sales, Scenario Profit, Scenario Profit Margin %, Scenario Profit Change	Scenario outputs
Clustered column chart	Actual Profit versus Scenario Profit	Baseline comparison
Variance visuals	Budget Sales, Sales Variance, Sales Variance % only when a real budget exists	Budget performance

Scenario DAX

```
Scenario Gross Sales = [Total Gross Sales] Scenario Discounts = [Scenario Gross Sales] * [Discount Rate Adjustment Value] Scenario Net Sales = [Scenario Gross Sales] - [Scenario Discounts] Scenario COGS = [Total COGS] * (1 + [COGS Change % Value]) Scenario Profit = [Scenario Net Sales] - [Scenario COGS] Scenario Profit Margin % = DIVIDE([Scenario Profit], [Scenario Net Sales], 0) Scenario Profit Change = [Scenario Profit] - [Total Profit]
```

Create the What-if parameters through Modeling → New parameter → Numeric range. Use the exact generated measure names in your model. The discount slider replaces the observed discount amount with an assumed rate of Gross Sales, so the scenario is a sensitivity analysis and may not equal actual profit at default settings.

5. Build, Test and Submit

Recommended implementation sequence

• Import the CSV and confirm the field names and data types.

• Create the date table, relationship, financial measures, and reconciliation checks.

• Build Page 1 first; then build Page 2 and Page 3 using the same design theme.

• Implement and validate the revenue waterfall.

• Create both What-if parameters and verify that the scenario cards and comparison chart respond to slider changes.

• Add budget variance only after receiving valid budget data at a compatible date/category level.

• Add navigation, synced slicers where appropriate, tooltips, concise titles, and consistent number formats.

• Save, close, reopen, and test the PBIX before submission.

Quality checklist

CHECK	PASS CONDITION
Data quality	Date and numeric columns have correct types; errors are investigated.
Financial arithmetic	Sales and Profit reconciliation checks pass or exceptions are documented.
Date behaviour	Months sort chronologically and date filters work.
Interactions	Country, Segment, Product and Year slicers filter intended visuals.
Scenario controls	Both sliders update the scenario cards and comparison chart.
Budget analysis	Targets are real/approved or clearly labelled illustrative assumptions.
Presentation	Consistent fonts, readable chart titles, suitable spacing and colour use.
Delivery	PBIX reopens correctly and contains all four pages.

Design recommendations

• Use a 16:9 page size and a consistent navy, blue, teal, and white theme.

• Place KPI cards near the top, trend charts in the middle, and detailed comparisons below.

• Use conditional colours carefully and keep backgrounds simple behind dense visuals.

• Apply consistent currency and percentage formatting across pages.

Expected deliverables

• **Corporate_Financial_Margin_Dashboard.pbix:** the four-page interactive Power BI report.

• **Cleaned CSV:** the source data used by the report.

• **Project report:** this PDF documenting design, data preparation, DAX, assumptions and testing.

• **Evidence:** screenshots of each completed page and a brief walkthrough, if requested.

Conclusion

The four-page design covers the main requirements in the internship brief: financial KPI monitoring, revenue erosion and margin analysis, segment/product comparisons, and interactive scenario modelling. The final report will be strongest when every measure is validated and when actual data is clearly separated from assumptions. Additional approved budget and cost-centre data can be incorporated if supplied by the internship mentor.

Reporting note: This document is an implementation guide. It does not claim the dashboard has already been built and does not invent financial results or budget figures.